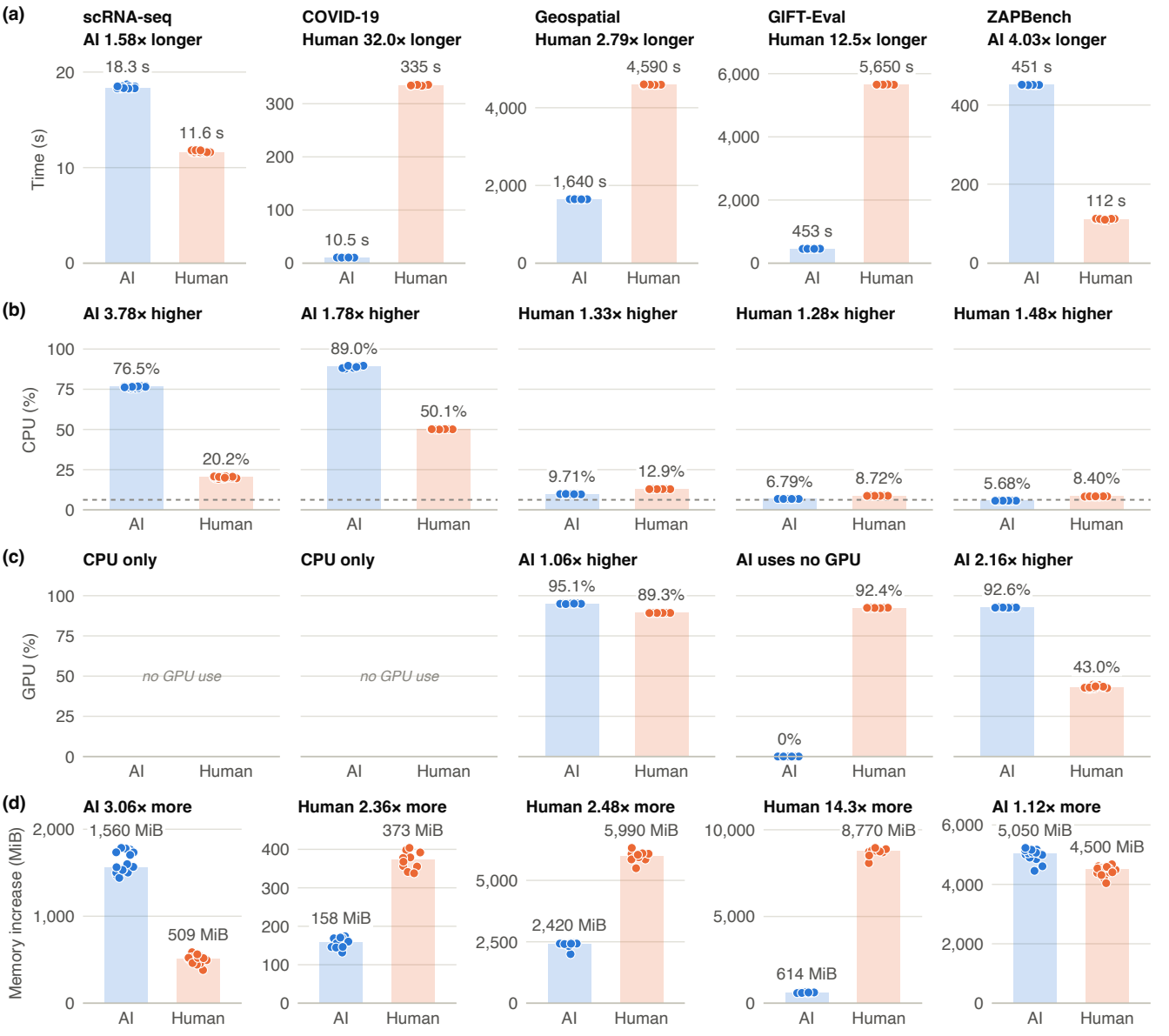

(a)
scRNA-seq
AI 1.58× longer
18.3 s
11.6 s
COVID-19
Human 32.0× longer
10.5 s
335 s
Geospatial
Human 2.79× longer
1,640 s
4,590 s
GIFT-Eval
Human 12.5× longer
453 s
5,650 s
ZAPBench
AI 4.03× longer
451 s
112 s
Time (s)
AI
Human
(b)
AI 3.78× higher
76.5%
20.2%
AI 1.78× higher
89.0%
50.1%
Human 1.33× higher
9.71%
12.9%
Human 1.28× higher
6.79%
8.72%
Human 1.48× higher
5.68%
8.40%
CPU (%)
(c)
CPU only
no GPU use
CPU only
no GPU use
AI 1.06× higher
95.1%
89.3%
AI uses no GPU
0%
92.4%
AI 2.16× higher
92.6%
43.0%
GPU (%)
(d)
AI 3.06× more
1,560 MiB
509 MiB
Human 2.36× more
158 MiB
373 MiB
Human 2.48× more
2,420 MiB
5,990 MiB
Human 14.3× more
614 MiB
8,770 MiB
AI 1.12× more
5,050 MiB
4,500 MiB
Memory increase (MiB)